

Hao Yuan

Postdoctoral Fellow

Department of Molecular Biosciences

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Google Scholar — Marcotte Lab — Personal website

Education

2021–2026	Ph.D. in Genetics and Genome Sciences; Ecology, Evolution and Behavior Topic: Contextualizing Biological Associations Across Genes, Cell Types, Species, and Evolution Advisor: Ingo Braasch, Arjun Krishnan Michigan State University, MI, USA
2016–2019	MS in Biology Topic: Assembly and Filtering of Enriched Data from Exon Capture Across Species Advisor: Chenhong Li Shanghai Ocean University, Shanghai, China
2012–2016	BS in Marine Biology Advisor: Chenhong Li Shanghai Ocean University, Shanghai, China

Professional Experience

2026–	Marcotte Lab, Postdoctoral Fellow
2021–2026	Braasch Lab & Krishnan Lab, Graduate Assistant
2020–2021	Shanghai Amplicongene Bioscience Co. LTD., Bioinformatician
2019	Genergy Biotechnology Co. LTD., Bioinformatician

Skills

Computational background and skills

- Well-versed with Python, R, Linux command line and version control using GitHub
- Experience with python package development
- Experience working with high-performance computing resources

Computational Genomics, Network Modeling, and Text Mining Skills

- Well-versed with genomic data analysis including genomic sequences, bulk and single-cell transcriptomics
- Experience with graph learning methods using protein-protein interaction network and integrated analysis with omic data
- Experience with statistical modeling using single cell omic data
- Experience with using large language models for large-scale text mining and metadata analysis

Statistics, Data science, and Machine learning background and skills

- Data wrangling and visualization in Python with pandas, matplotlib, and seaborn
- Statistical modeling and machine learning in Python with numpy, numba, statsmodel, scipy, scikit-learn and PyTorch

Publications

[* Joint Primary Author]

- 12. Large-scale automated detection reveals pervasive sex imbalance in biomedical research.**
Valtadoros, L., Hicks, P., Yuan, H., Ahmadianm M., Johnson, K., & Krishnan, A.
bioRxiv (2026), 07.13.738332.
<https://doi.org/10.64898/2026.07.13.738332>
- 11. Linking Genetic Risk to Disease-Relevant Cellular States via Metacell-Informed Modeling with ICePop.**
Yuan, H., Mandava, A., Sarmart, K., Ganz, J., & Krishnan, A.
bioRxiv (2026), 04.01.715877.
<https://doi.org/10.64898/2026.04.01.715877>
- 10. Computational strategies for cross-species knowledge transfer and translational biomedicine.**
Yuan, H., Mancuso, C. A., Johnson, K., Braasch, I., & Krishnan, A.
Nature Methods (2025): 1–16.
<https://doi.org/10.1038/s41592-025-02931-9>
- 9. Annotating publicly-available samples and studies using interpretable modeling of unstructured metadata.**
Yuan, H., Hicks, P., Ahmadian, M., Johnson, K. A., Valtadoros, L., & Krishnan, A.
Briefings in Bioinformatics (2024), 26(1).
<https://doi.org/10.1093/bib/bbae652>
- 8. Estimation of Species Abundance Based on the Number of Segregating Sites using Environmental DNA (eDNA).**
Ai, Q.*, Yuan, H.*, Wang, Y., Li, C.
Molecular Ecology Resources (2025), 25(6), e14076.
<https://doi.org/10.1111/1755-0998.14076>
- 7. CONE: COntext-specific Network Embedding via Contextualized Graph Attention.**
Liu, R.*, Yuan, H.*, Johnson, K. A., & Krishnan, A.
Proceedings of Machine Learning Research (2024), 261:53–71.
<https://proceedings.mlr.press/v261/liu24a.html>
- 6. Exon-capture data and locus screening provide new insights into the phylogeny of flatfishes (Pleuronectoidei).**
Atta, C. J., Yuan, H., Li, C., Arcila, D., Betancur-R, R., Hughes, L. C., Ortí, G., & Tornabene, L.
Molecular Phylogenetics and Evolution (2022), 166, 107315.
<https://doi.org/10.1016/j.ympev.2021.107315>
- 5. Inline index helped in cleaning up data contamination generated during library preparation and the subsequent steps.**
Wang, Y., Yuan, H., Huang, J., & Li, C.

Molecular Biology Reports (2022), 49(1), 385–392.
<https://doi.org/10.1007/s11033-021-06884-y>

4. **Assexon: assembling exon using gene capture data.**
Yuan, H., Atta, C., Tornabene, L., & Li, C.
Evolutionary Bioinformatics (2019), 15, 1176934319874792.
<https://doi.org/10.1177/1176934319874792>
3. **Gene markers for exon capture and phylogenomics in ray-finned fishes.**
Jiang, J. *, Yuan, H. *, Zheng, X., Wang, Q., Kuang, T., Li, J., Liu, J., Song, S., Wang, W., Cheng, F., Li, H., Huang, J., & Li, C.
Ecology and Evolution (2019), 9(7), 3973–3983.
<https://doi.org/10.1002/ece3.5026>
2. **A phylogenomic approach to reconstruct interrelationships of main clupecocephalan lineages with a critical discussion of morphological apomorphies.**
Straube, N., Li, C., Mertzen, M., Yuan, H., & Moritz, T.
BMC Evolutionary Biology (2018), 18(1), 158.
<https://doi.org/10.1186/s12862-018-1267-1>
1. **Target gene enrichment in the cyclophyllidean cestodes, the most diverse group of tapeworms.**
Yuan, H., Jiang, J., Jiménez, F. A., Hoberg, E. P., Cook, J. A., Galbreath, K. E., & Li, C.
Molecular Ecology Resources (2016), 16(5), 1095–1106.
<https://doi.org/10.1111/1755-0998.12532>

Conference and Meeting Presentations

Talks

Outside MSU

- **Sep 2024**
CONE: COnText-specific Network Embedding via Contextualized Graph Attention.
Machine Learning in Computational Biology 2024, Seattle, WA
- **Mar 2024**
Discovering context-specific functionally equivalent genes in research organisms using cross species transcriptome-based machine learning.
The Allied Genetics Conference 2024, Metro Washington, DC
- **Dec 2022**
An ML framework for precision medicine: from patient-specific gene networks to translational animal models.
Rocky 2022 Bioinformatics Conference, Snowmass, CO
- **Oct 2016**
Genome-wide detection of sites under selection using a modified method of evolutionary probability — a case study on adaptive evolution of Homo species.
23rd Academic Annual Meeting of China Zoological Society, Wuhan, China

Within MSU

Genetics and Genome Sciences Program

- **Oct 2024**
Annotating public omics samples and studies using interpretable modeling of unstructured metadata by txt2onto 2.0.
Genetics and Genome Sciences Forum
- **Feb 2024**
An ML framework for precision medicine: from patient-specific gene networks to translational research organisms.
Genetics and Genome Sciences Forum
- **Oct 2023**
Reconstruct patient-specific gene networks from transcriptomic data.
Genetics and Genome Sciences Forum

Computational Mathematics, Science and Engineering Program

- **Oct 2022**
An ML framework for precision medicine: from patient-specific gene networks to translational animal models.
Computational Biology Forum

Posters

Outside MSU

- **2025**
Evolutionary Implications of the Teleost Genome Duplication Revealed by Cross-Species Single-Cell Transcriptomic Comparison.
American Society for Biochemistry and Molecular Biology 2025: Evolution and Core Processes in Gene Expression, Kansas City, MO
- **2025**
ICePop: Identifying Disease-Affected Cell Types Through Network-Based Analysis of Gene Module Overexpression in Single-Cell Data.
Network Biology 2025, Cold Spring Harbor, NY
- **2024**
ICePop: Identifying Disease-Affected Cell Types Through Network-Based Analysis of Gene Module Overexpression in Single-Cell Data.
Biological Data Science 2024, Cold Spring Harbor, NY
- **2024**
Tracking dynamic regulatory changes in rare diseases using single sample networks.
Annual Meeting of American Society of Human Genetics 2024, Denver, CO
- **2024**
CONE: CONtext-specific Network Embedding via Contextualized Graph Attention.
Machine Learning in Computational Biology 2024, Seattle, WA
- **2023**
Interpretable text-based machine learning for inferring systematic tissue and disease annotations of public transcriptome samples.
Genome Informatics 2023, Cold Spring Harbor, NY

- **2023**
An ML framework for precision medicine: from patient-specific gene networks to translational animal models.
Network Biology 2023, Cold Spring Harbor, NY
- **2022**
An ML framework for precision medicine: from patient-specific gene networks to translational animal models.
Rocky 2022 Bioinformatics Conference, Snowmass, CO
- **2022**
Cross-species transcriptome-based regression to discover equivalents of human samples and genes in biomedical research organisms.
30th Conference on Intelligent Systems for Molecular Biology, Madison, WI
- **2018**
EXpipe: An Assembly Pipeline for Exon Capture Data Across Large Scales of Divergence.
1st AsiaEvo Conference, Shenzhen, China

Honors and Awards

- 2026 Genetics and Genome Sciences Program Outstanding Student Award
- 2023 Summer College of Natural Science Outstanding Scholar Fellowship

Teaching and Mentoring Experience

Teaching

- **2023** IBIO341: Fundamental Genetics, Teaching Assistant

Mentoring

Post-baccalaureate

- **2023–2025** Lydia Valtadoros, Department of Biomedical Informatics, The University of Colorado Anschutz Medical Campus

Junior graduate student

- **2023–2026** Parker Hicks, Department of Biomedical Informatics, The University of Colorado Anschutz Medical Campus

Professional Service

Reviewing Service

- **2026** *G3: Genes—Genomes—Genetics* (2 manuscripts)
- **2025** *G3: Genes—Genomes—Genetics* (1 manuscript)
- **2024** *Journal of Experimental Zoology Part B: Molecular and Developmental Evolution* (1 manuscript co-review with Dr. Ingo Braasch)
- **2023** *Genome Biology* (1 manuscript co-review with Dr. Arjun Krishnan)

- **2022** *Genome Research*
(1 manuscript co-review with Dr. Arjun Krishnan)

Community Service

- **2023–2025** Genetics and Genome Sciences Program Seminar Coordinator
- **2023–2025** Genetics and Genome Sciences Program Graduate Student Organization

Career Development Activities

- **Jan 2023** Graduate Student Career Forum
- **Apr 2022** APIDA Virtual Career Panel: Careers in Data, Analytics, Technology, & Engineering
- **Apr 2022** Boost your professional power skills

STEM Outreach

- **Dec 2023** AP Biology DeWitt High School Visit Day